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HPC research project update

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Fri, Nov 13, 2020 at 3:55 PM

To: Mitali Sinha <mitalis@iiitd.ac.in>

Cc: Fahad Nayyar <fahad17049@iiitd.ac.in>, Vishal Samal <vishal17209@iiitd.ac.in>, Sujay Deb <sdeb@iiitd.ac.in>

Sir,

This is to inform that we got DATE 2021 results today evening for our HPC work. Though the paper missed acceptance, the idea has got good feedback as far as we could see. Following attachment is the mail received from them.

Thanks and Regards,

FEEDBACK RECEIVED:

On behalf of the Technical Program Committee, I am sorry to inform you that your submission to DATE 2021

Title: HPC-Boost: A Binary driven ML Framework for boosting HPC based Runtime Attack Detection

Authors: M Badri, Fahad Nayyar, Vishal Samal, Mitali Sinha and Sujay Deb

Number: #1970

has not been accepted.

The selection process was very competitive. Indeed, the acceptance rate of regular papers is 24%. Cumulative acceptance rate for all regular papers and IPs is 36% of the submitted papers. Submissions involved more than 2600 people, from 33 different countries.

The reviews related to your submission from the DATE TPC committee are included at the end of this e-mail. I hope they will provide you with useful feedback. Please note that this year the overall score ranges from -3 to +3.

I would like to sincerely thank you for your submission to DATE 2021 and, despite the outcome of this submission, I hope to see you at the online DATE 2021 event during February 1-5, 2021.

The full conference program (including your paper slot and allocated IP session) will be published online very soon at <https://www.date-conference.com>.

Please note that the registration to the conference is only possible via the online registration platform, which is now open: <https://www.date-conference.com/registration>

Best regards,

Ian O'Connor, TPC Chair DATE 2021
DATE 2021 Program Chair

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<http://www.date-conference.com>

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DATE 2021 Reviews for Submission #1970

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Title: HPC-Boost: A Binary driven ML Framework for boosting HPC based Runtime Attack Detection
Authors: M Badri, Fahad Nayyar, Vishal Samal, Mitali Sinha and Sujay Deb

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REVIEWER #1

Reviewer's Scores

Overall Value: 1

Summary of ideas

This paper works in the context of anomaly detection using Hardware Performance Counters (HPCs), whose values over time can be used to characterize benign and malicious programs. Here the authors consider the problem of selecting which counters to use for optimal performance - Although many (400-500) exist, only few registers (4 for Intel, 6 for AMD) can be used to sample them at the same time. Although more can be collected at the same time, this will be done in a multiplexed fashion. Therefore, the authors of this paper discuss how to select top HPCs for a given platform. The purpose of this selection is to classify binaries as malware or benign programs. Therefore, the approach consists of (roughly) two steps: 1) a "feature extractor" that extracts features from a binary and 2) a "ranking module" that classifies the extracted features with a trained ML model.

Strong points

Besides a number of typos, the paper is well written and easy to follow. The problem under discussion is an interesting one, as it is an obvious selection problem that is so far (mostly) ignored in the literature. The approach of selection is interesting, capturing various important aspects of the counters (e.g., consistency, variability, inter-correlation). By using a feature extractor there is no need for a large data collection at each deployment, which improves the ease-of-use. On the flipside, this creates an additional approximation layer that could lose some precision. I expect that this is worth it, as even slightly suboptimal events will lead to a performance improvements over poorly chosen counters.

Weak points

The description of the malicious programs could be extended, as it is not very clear how significant the modifications to benign programs are. Do the results change at all depending on the type of malware that is compared against? For example, if branch instructions are included then the branch-related counters should have an advantage in detection of such malware - Is this taken into consideration at all?

Other comments

- * p1: "boosting" in title should be capitalized?
- * p1: "Due to its ability" -> "Due to their ability"
- * p1: "to unreliable" -> "to an unreliable"
- * p1: "almost 4" -> "at most 4"
- * p1: "of actual binary" -> "of the actual binary"
- * p1: "train ML model" -> "train a ML model"
- * p1: "ml model" -> "ML model"
- * p2: Various sentences starting with a reference "[...]", which is bad form
- * p3: The various parts of the ranking module are first listed and then discussed in a different order, which is slightly confusing
- * p3: The "Hirsh et al., 1991" reference is in the wrong style?
- * p3: I would suggest mentioning the normalization of the scores before discussing the different scores (it is now remarked afterwards), as this confused me initially in the description
- * p4: "long100 and beyond" missing bracket
- * p4: "They [21]" weird use of the word "They"
- * p5: "This neural network learn" -> "This neural network learns"

REVIEWER #2

Reviewer's Scores

Overall Value: -2

Summary of ideas

The main idea of the paper is to use HPC events to detect runtime attacks. The authors proposed to monitor selective HPC events because of constraints in the number of registers in a processor. The HPC events are selected based on the statistical ranking proposed by the authors. In addition, the authors also used machine learning for mapping features of a binary program to the ranked HPC events to boost the performance. A recommender system is also used to provide suitable HPC events for a particular binary such that the performance of attack detection increases.

Strong points

The idea of ranking HPC events for a particular binary program is nice.

Weak points

There is little to no contribution.

The paper is technically not strong. Some details are mentioned superficially or omitted intentionally.

Experiments are not validated using any standard set of attack programs. Instead, the authors used self-crafted attack programs.

There is no discussion or validation on the overhead of the methodology.

Other comments

The editorial quality of the paper is very poor. There are a significant number of grammatical and typographical errors in the manuscript. Most of the figures and tables are not explained at all, which makes the paper difficult to follow. However, these are not considered during the technical evaluation of the manuscript. The paper has several technical issues, which are listed below:

1. "We use this mechanism to create a labeled dataset of workload.." - Does the labeled dataset contain both malicious and benign processes? It is not clearly mentioned. I assume, since the objective of the paper is to detect an anomaly, the dataset contains only benign workloads. Then, what are the types of workload used for dataset creation? For which features of the unknown attack programs, the method will detect it as an anomaly? The motivation behind the correctness of the proposed approach is missing from the paper.
2. "Different binaries should be monitored with different sets of HPC events for better anomaly detection" - For an attack program during runtime, how the method works? I understand that the HPC events are selected by analyzing the HPC values and binary features from the recommender system. But to get any statistically significant HPC observations, any program needs to be executed multiple times or for a specific window. What is the detection time of the attack? Do they get detected before its completion? For example, in Fig. 4 the x-axis is time step in minutes. Most of the well-known attacks (malwares, trojans, etc.) do not require such an amount of time.
3. The authors have mentioned performing time-series analysis. Then I assume the HPC events are sampled. What is the sampling frequency? Different binaries can behave differently for a fixed sampling frequency. How the methodology handles such an obstacle?
4. Where the time-series component is considered in the equation on Page 3? First of all, the equation is not explained properly, and it does not have any expression to handle the time component.
5. The discussion on the stationary score is a bit verbose. It will be better to understand if it is explained with some mathematical expressions.
"Depending on the length of the time series data at hand, we classified the data streams as short(25-100) and long100 and beyond)." - The idea of short and long is not clear in the discussion.
6. What is the execution time of the feature extractor module? How it affects the detection time

during runtime?

7. Providing results on self-generated attack codes is unfair for the evaluation. The authors should consider actual malware codes. There are lots of repositories available on the internet for such programs.

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REVIEWER #3

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Reviewer's Scores

Overall Value: -1

Summary of ideas

The paper proposes a new event recommendation framework to enhance the run-time detection of anomalies using HPC.

The framework consists of 3 main phases:

- feature extractor,
- ranking module,
- recommender module.

An experimental study is provided on an intel i5, Ubuntu 18.04 LTS, where anomalies are artificially added by changing opcodes, operands, insert function calls, and branch/jmp instructions.

Strong points

The topic of the paper and their idea of giving information on the relevance of HPC is of great interest to the community.

Weak points

Detailed proofreading would greatly benefit the quality of the paper.

The experimental study/experimental setup is not detailed enough, especially as the binaries are self-made and HPC events/importance highly depend on the underlying activities.

Other comments

The results (highlighted in B.2) and B.3)) are kind of straightforward and more insights would improve the quality of the paper. For example, dividing the binary set into different categories and studying the differences in HPC events.

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REVIEWER #4

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Reviewer's Scores

Overall Value: -1

Summary of ideas

While state-of-the-art runtime attack detection tools using HPC don't differentiate the events set, and their importance for the decisional criteria, the authors propose a "binary features-driven HPC event recommendation framework" to improve runtime attack detection.

Indeed, the authors noticed that HPC based anomaly detectors rarely pay attention to the properties of actual binary under attack. Moreover, the new idea includes to use statistical methods to rank the events based on the sampled HPC data streams of the programs.

The framework is mainly composed by three modules. For each binary, the "Ranking module" monitors every HPC event. Then, the "Feature Extractor module" extracts relevant binary features in the form of a vector. The goal of the Feature Extractor is to capture the essence of the binary in a lightweight manner. Then, during the training time, the model in the "Recommender module" trains on these binary's feature vectors and probability scores corresponding to the HPC event ranking for that binary. Finally, during the testing phase, the Feature extraction module extracts the feature vector from the new binary, and the trained Recommender module obtains the HPC event ranking for this feature vector. Tests show that binary specific event selection is important, because exchanging different ranking position events causes efficiency degradation.

Strong points

The idea is interesting, and it seems to work.
The paper also addresses false positive and negatives, improving the efficiency of the framework, and showing that some deep work on the topic has been done.
In terms of performance, the pipeline composed by the three modules won't incur much overhead, because authors claim they are not computationally heavy, once deployed after the training.

Weak points

It would be interesting to see more data, especially a comparison between the proposed solution and previous solutions, both in terms of performance and efficiency. Moreover, data and charts in the paper are not easily readable, and often units of measures are missing and poor explanations are given.

Other comments

page 3, section B, paragraph "Spread Score": "in the of": missing word?
same page, second columns: "This test however independent of the time series nature of the data though": missing verb?
fig. 5 labels (a) and (b) are not clear
poor acronyms explanations (e.g. CFG, BFS, TP, TN, ...), probably given as known?
figure 6 misses legend

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DATE 2021 - <https://www.softconf.com/date21/conference>

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